Final Project Progress Update

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GitHub repository: https://github.com/kwoolaid725/msds-457 (package in `daimondstats/`)

Completed Work

daimondstats version 0.0.3 works with scorecard exports from dAImond Inning. import_scorecards() loads and validates an export. Three functions provide the analysis:

- `compare_personal_season()` compares the games a user scored with the team's other games that season. It draws random sets of the same size that keep the same home/away mix, then reports where the user's games fall among them.
- `calculate_run_impact()` calculates RE24 for each play: runs scored plus the ending run expectancy, minus the starting run expectancy (Weinberg 2014).
- calculate_batting_stats() returns batting totals, batting average, on-base percentage, slugging percentage, and OPS by game, team, and player. It rejects entries with unresolved review flags.

The package has DESCRIPTION, NAMESPACE, README, roxygen documentation for its public functions, three documented datasets, and base-R tests (Wickham and Bryan 2023). The new reviewed_scorecards dataset contains 179 plate appearances from five team scorecards, covering 49 player/game combinations. I checked every event against a live MLB feed and compared all player and team counting totals with the official box scores. Ten exported labels were normalized; original labels and review notes are retained. R CMD check passed with status OK, including examples and all three test scripts.

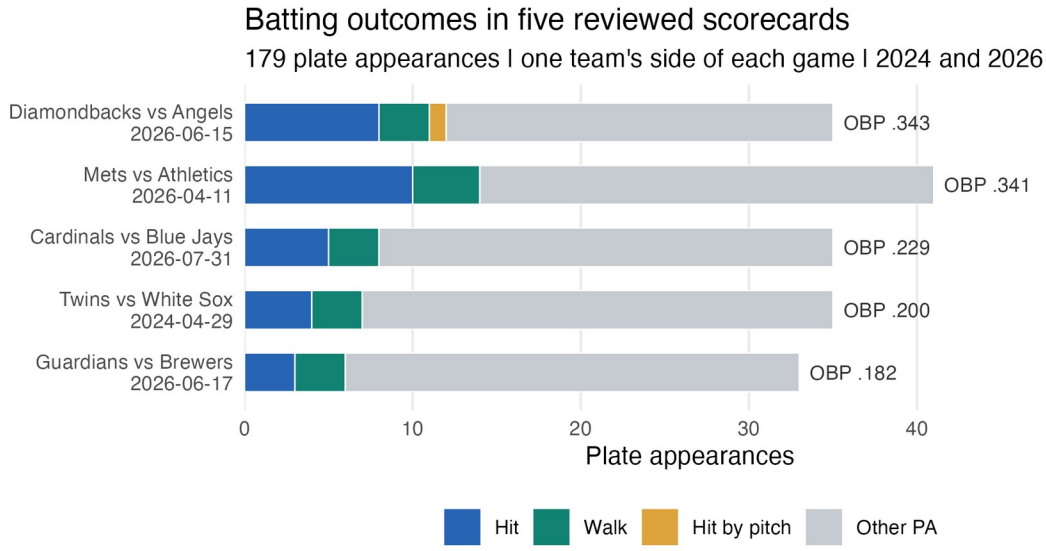


Figure 1. Plate appearance results and OBP for five reviewed team scorecards, calculated with calculate_batting_stats() from reviewed_scorecards. Other PA includes outs, errors, fielder choices, and a sacrifice fly. Sources: saved dAImond Inning cards and MLB feeds checked October 25, 2026 (Major League Baseball 2026).

OBP ranges from .182 for Cleveland to .343 for Arizona. Arizona and the Mets have similar OBPs (.343 and .341), although the Mets recorded ten hits in 41 plate appearances and Arizona eight in 35. These are five recorded team games across two seasons, so the chart does not establish which team is better.

Remaining Work and Plan

The season comparison and RE24 have only been tested with made-up inputs. I plan to finish in three steps:

1. **Reference data.** Add a complete team season and link the reviewed games through MLB game IDs. For RE24, obtain verified before/after base-out states and a documented run expectancy table. The current dataset does not contain those states.
2. **Analytical validation.** The batting workflow now matches official box scores. Next, run the personal-season comparison on a coherent team-season reference and check several real-event RE24 values by hand. Add tests for any discrepancies.
3. **Final materials.** Write a vignette that goes from the app's export to analysis and a plot. Then finish the technical documentation, the how-to guide with three walkthroughs, and the presentation. Before submitting, I will install from GitHub and rerun the package checks.

Challenges

Data quality. I audited 155 saved results, including child cards inside multi-card uploads. Of eleven component cards with event references, five passed the full review. I excluded extra detections, incomplete coverage, duplicate runs, and an unresolved single-versus-error conflict. Corrections are recorded rather than silently replacing the source labels.

Missing game situation. A play label like "single" does not say where the runners went, and RE24 needs that. I will use verified event records instead of guessing runner movement from the play label.

Selection bias. These five scorecards cover different teams and seasons and form a convenience sample. They are useful for checking the workflow, but they are not a season benchmark. I will use a coherent reference season for the personal-season comparison and treat its results as descriptive.

Event definitions. The cached reference classified force outs as groundouts. I checked whether the batter reached first in the MLB description and used fielder choice where appropriate. Matching the cached label alone would have missed that distinction.

References

Major League Baseball. 2026. "MLB Stats API Game Feeds." Accessed October 25, 2026. https://statsapi.mlb.com/api/v1.1/game/746804/feed/live. Per-game URLs are recorded in the package verification manifest.

Weinberg, Neil. 2014. "RE24." FanGraphs, July 30. https://library.fangraphs.com/misc/re24/.

Wickham, Hadley, and Jennifer Bryan. 2023. *R Packages*. 2nd ed. Sebastopol, CA: O'Reilly Media. https://r-pkgs.org/.